

# Fall Internship Project

## ExaTune: Exhaustive Hyperparameter Landscape Exploration via HPC for Machine Learning Models

BORN Ontario, Sept.-Dec. 2025

Intern: Katie Lonsway  
Supervisor(s): Kevin Dick, Heather Howley

### 1. Project Overview

Hyperparameter optimization is a cornerstone of machine learning practice. Conventional techniques such as grid search, random search, Bayesian optimization, and genetic algorithms balance computational efficiency against exhaustive search. However, these methods often trade completeness for speed, potentially overlooking regions of the hyperparameter landscape that reveal insights into model stability, chaotic regions of parameter sensitivity, or the presence of multiple performance optima.

With the rise in availability of high-performance computing (HPC) infrastructure, particularly those orchestrated via SLURM, there exists growing opportunity to revisit exhaustive exploration at scale. By distributing each hyperparameter configuration as an independent job, one can map the entire hyperparameter performance landscape, given sufficient compute resources and generate comprehensive performance evaluation. The results can then be studied as a landscape: smooth regions, sharp cliffs, and chaotic pockets of instability.

For applied healthcare research projects, the development of a robust hyperparameter experimentation framework will facilitate the development of highly accurate models by systematically uncovering optimal parameter configurations, characterizing regions of stability, and revealing sensitivities that may affect model generalizability. Such insights are critical in healthcare contexts where small improvements in predictive performance can translate into significant clinical impact, such as earlier disease detection of rare conditions, more precise risk stratification, or improved allocation of limited resources. Beyond accuracy, a comprehensive view of the hyperparameter landscape also strengthens the reproducibility and interpretability of research findings, ensuring that models deployed in clinical or policy settings are not only performant but also trustworthy, transparent, and resilient across diverse patient populations.

A generalized, reusable, pip-installable Python package that integrates HPC job distribution, standardized ML models (e.g., scikit-learn, XGBoost), and rich visualization will provide both practical utility and novel research contributions. The accompanying research article will analyze emergent patterns in hyperparameter landscapes, contributing to both ML methodology and HPC systems research.

### 2. Project Objectives

The following project comprises three main objectives: (i) the development of the *exatune* Python package to generate exhaustive performance landscapes, (ii) the extension of the *exatune* package to support various forms of visualization, (iii) and the application of this project to various benchmark datasets and summarization of research findings in a publication.

1. **Develop a Python package** (exatune) that:

- Accepts any ML model from scikit-learn or XGBoost.
- Generates all hyperparameter combinations from a configuration file.
- Distributes experiments across arbitrary SLURM HPC resources.
- Collects results into a structured database/JSON/Parquet format.
- Provides APIs for visualization (2D heatmaps, 3D surfaces, interactive projections for higher dimensions).

2. **Visualization Analysis Toolkit:**

- Traditional heatmaps/surfaces for 2D–3D parameter pairs.
- Dimensionality reduction (PCA, t-SNE, UMAP) for higher dimensions.
- Interactive dashboards (e.g., Plotly or Bokeh).

3. **Research Study:**

- Apply the tool to benchmark ML tasks (classification and regression).
- Characterize landscapes: smooth vs chaotic, multimodality, parameter interactions.
- Publish findings on landscape structures and implications for optimization.

### 3. Project Phases & Tasks

The sections below outline the specific monthly tasks and deliverables towards successfully completing this internship project.

#### Phase 1: Foundations (Weeks 1–4)

- Task 1.1: Literature review on hyperparameter optimization, landscape analysis, and HPC scheduling.
- Task 1.2: Design package architecture:
  - CLI & Python API
  - Configuration-driven experiment definitions (YAML/JSON)
  - SLURM job submission & monitoring interface
- **Deliverable:** Package skeleton (`exatune`) with config parsing and job stub generator.

#### Phase 2: HPC Integration (Weeks 5–8)

- Task 2.1: Implement SLURM job submission, monitoring, and result collection.
- Task 2.2: Support scikit-learn and XGBoost model wrappers with fit/evaluate routines.
- Task 2.3: Test scalability across varying SLURM cluster sizes.
- **Deliverable:** End-to-end pipeline running exhaustive search on toy datasets.

#### Phase 3: Visualization & Analysis (Weeks 9–10)

- Task 3.1: Implement plotting APIs: heatmaps, 3D surfaces, interactive visualizations.
- Task 3.2: Implement higher-dimensional reduction for landscape exploration.
- Task 3.3: Build an interactive dashboard module for exploration.
- **Deliverable:** Visualization library integrated with results.

#### Phase 4: Research Application (Weeks 11–12)

- Task 4.1: Run full experiments on benchmark datasets (e.g., MNIST, UCI datasets).

- Task 4.2: Characterize landscapes quantitatively: variance across neighbors, multi-modality, chaotic sensitivity.
- Task 4.3: Draft internal report and visual appendix.
- **Deliverable:** Research findings and draft manuscript sections.

### Phase 5: Paper Drafting & Finalization (Weeks 13–16)

- Task 5.1: Write research article: background, methods, software description, results, implications.
- Task 5.2: Iterate with supervisors on analysis depth and framing.
- Task 5.3: Finalize software documentation and PyPI release.
- **Deliverables:**
  - Submitted manuscript to relevant venue (e.g., *Journal of Machine Learning Research: Open Source Software*, or *ACM Transactions on HPC*).
  - Public PyPI package with documentation and GitHub repository.

## 4. Computing & Research Environment

The code development for this project can be completed on the BORN Ontario laptop however the testing and research component of this work require access to a SLURM HPC cluster; to this end we will use the Digital Research Alliance of Canada (DRAC; formerly "compute canada") HPC resources within this project.

Access to the server is possible by applying for an account and selecting a role as an undergraduate researcher under Dr. Kevin Dick's research group.

## 5. Tools & Libraries

- **Core ML:** scikit-learn, XGBoost
- **HPC Integration:** SLURM, joblib, Jinja2 (for sbatch templates)
- **Visualization:** matplotlib, plotly, bokeh, umap-learn, PCA/t-SNE
- **Data Handling:** pandas, pyarrow, numpy
- **Configuration & Parsing:** pydantic, yaml, json
- **CLI & Interfaces:** click/typer, rich (for logging/CLI output)
- **Documentation:** LaTeX, MkDocs, ReadTheDocs
- **Version Control:** Git + GitHub (CI/CD via GitHub Actions)

## 6. Contact and Supervision

Project supervision will include weekly virtual (Teams) check-ins, review of code contributions via GitHub, and optional support for presenting the work at a local or virtual symposium.

The intern is expected to attend the weekly update meetings prepared with a detailed summary of the work completed during the previous week, along with a clear plan outlining their objectives and tasks for the upcoming week. These updates must be presented in a standardized written format, including sections for accomplishments, challenges encountered,

time spent, and proposed next steps. This reporting structure is intended to ensure transparency, maintain steady progress, and facilitate timely feedback. Weekly reports will serve as the primary tool for tracking project milestones, identifying bottlenecks, and adjusting the project scope or timeline as needed.